
unseen dataset
metadata + clock
low-budget fallback:
under 35 min: skip stages 2, 4, 5 (encoded -> CNN default)
under 15 min: also skip stage 3
1 MEASURE
fingerprint: neighbouring-pixel correlation (corr1)
on the raw inputs, seconds of CPU, no training
2 BACKBONE CHOICE family in {CNN, 1-D TCN, 2-D TCN}
encoded data?
(corr1 < 0.2)
yes: encoded
no: image-like
skip the bake-off
backbone := 2-D TCN
(channel-mixing stem if >= 8 channels)
saved time goes to training
BACKBONE GATE
train each family briefly
at matched wall-clock
leader clearly beats
the CNN?
(margin > noise)
yes
no
backbone := leader
backbone := CNN
(encoded default)
3 AUGMENTATION PROBE after the backbone choice, matched to the chosen family
two small networks OF THE CHOSEN FAMILY,
trained briefly: with vs without augmentation
does OFF win by a clear margin?
(asymmetric rule: augmentation is
regularisation, so OFF leads early)
yes
no
aug := off + mixup
aug := on (the default)
4 DEPTH SEARCH depth in {4, 6, 8}, default 6
train depth-4 and depth-8 candidates briefly against
the INCUMBENT default depth 6 (skipped for the 1-D TCN)
candidate beats the incumbent
by > noise AND survives
re-training verification?
yes
no
adopt searched depth
keep default depth 6
5 RANDOM SEARCH ARM random architectures: 11 operation types, depth 3 to 12, width 16 to 48
encoded AND
budget remains?
yes
sample 6 random architectures,
train each briefly
clearly better than
the incumbent?
(margin > 2x noise)
no budget / image-like
yes
no
adopt random architecture
discard, keep incumbent
6 TRAIN TO BUDGET
cosine learning rate annealed to zero, keep the best checkpoint
predictions